

Olympiad Test Paper — Science (Class 7)

Chapter 14: Electric Current and its Effect

Subject	Science (NCERT Class 7)
Chapter	14 — Electric Current and its Effect
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Recall the circuit symbols (cell = one long thin line + one short thick line; battery = two or more such cells in a row; bulb = a circle with a cross inside; closed switch = a closed gap, open switch = a raised arm leaving a gap; wire = a plain line). A bulb glows only when the circuit is a complete (closed) loop. No calculator is needed. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

1. In the symbol for a single electric cell, the **longer thin line** and the **shorter thick line** stand for, respectively:

- (A) the positive terminal and the negative terminal
- (B) the negative terminal and the positive terminal
- (C) the wire and the switch
- (D) two equal terminals with no fixed sign

2. A battery is correctly built from three identical cells by joining them so that:

- (A) the positive terminal of one cell touches the positive terminal of the next
- (B) all three negative terminals are joined at one point
- (C) the positive terminal of one cell joins the negative terminal of the next, in a row
- (D) the cells are stacked with a gap of air between each pair

3. Two identical cells are placed in a torch the **wrong way** — positive terminal facing positive terminal in the middle. The bulb will:

- (A) glow twice as bright
- (B) glow at normal brightness
- (C) not glow, because the cells oppose each other
- (D) glow, but only half as bright

4. A circuit has one cell and one bulb, and the bulb glows dimly. A student replaces the single cell with a battery of two cells joined end-to-end (+ to -) the right way. The bulb now:

- (A) glows more brightly
(C) goes out completely
- (B) glows more dimly
(D) glows exactly as before

5. The symbol drawn as a **small circle with a cross (×) inside it** represents a:

- (A) cell
(C) bulb
- (B) switch
(D) fuse

6. Which circuit symbol is drawn simply as a **plain straight line**?

- (A) a cell
(C) a bulb
- (B) a connecting wire
(D) an open switch

7. In a circuit diagram an electrician draws a small gap in the line with a tilted arm lifted away from one contact. This symbol means the switch is:

- (A) closed, and current flows
(C) a fuse that has melted
- (B) open, and current does not flow
(D) a battery being charged

8. A simple circuit is wired as: cell → wire → bulb → wire → **open switch** → back to cell. With the switch in this state the bulb is:

- (A) glowing brightly
(C) not glowing
- (B) glowing dimly
(D) glowing and then slowly fading

9. A torch bulb does **not** light when its switch is pressed. Which of these is **not** a possible reason?

- (A) one of the connecting wires is broken
(C) the circuit is a complete closed loop
- (B) the cell is dead
(D) the bulb's filament is fused

10. For a bulb to glow in a circuit, the essential condition is that:

- (A) the circuit must be open at one point
(B) the circuit must form a complete closed loop
(C) there must be at least three cells
- (D) the switch must be open

11. A battery is made of four identical 1.5 V cells joined correctly in a row (+ to −). The total voltage pushing current round the circuit is about:

- (A) 1.5 V
(C) 6 V
- (B) 0.375 V
(D) 4 V

12. Two pieces of wire, P and Q, are each connected in turn across the same cell with the same bulb. With P the bulb is bright; with Q the bulb is dim. The most likely explanation is that:

- (A) wire Q has a higher resistance than wire P
(C) wire P is broken inside
- (B) wire Q is a perfect insulator
(D) wire Q has no resistance at all

13. Which of the following is the best **conductor** of electric current, and would let a bulb glow if used as the connecting link?

- (A) a dry wooden stick
- (B) a rubber band
- (C) a copper wire
- (D) a plastic ruler

14. A student joins a bulb to a cell using a strip of **dry rubber** in place of one wire. The bulb does not glow because rubber is:

- (A) a good conductor
- (B) a low-melting-point metal
- (C) magnetic
- (D) an insulator, so the loop is not completed electrically

15. Electric current passing through a wire always makes the wire:

- (A) cooler than the surroundings
- (B) glow with light, whatever the wire
- (C) warmer, because of the heating effect of current
- (D) lose all of its resistance

16. Two wires of the **same length and material** carry the same current. Wire X is thin and has high resistance; wire Y is thick and has low resistance. Compared with Y, wire X gets:

- (A) cooler
- (B) exactly as hot
- (C) hotter
- (D) magnetised but not heated

17. The coil inside an electric room heater is made of **nichrome** rather than copper because nichrome:

- (A) has a very high resistance and a high melting point, so it heats up and glows without melting
- (B) is a much better conductor than copper
- (C) is magnetic
- (D) is cheaper than any other metal

18. As the current through a given heating wire is increased, the amount of heat produced in the wire each second:

- (A) decreases
- (B) becomes zero
- (C) stays exactly the same
- (D) increases

19. The filament of an electric bulb glows white-hot when current flows. It is made of **tungsten** mainly because tungsten:

- (A) is a magnetic metal
- (B) has an extremely high melting point, so it can get white-hot without melting
- (C) is an insulator
- (D) melts very easily at a low temperature

20. Which set of appliances **all** work mainly on the **heating effect** of electric current?

- (A) electric iron, toaster, immersion water heater
- (B) electric bell, crane magnet, loudspeaker
- (C) ceiling fan, mixer, water pump
- (D) compass, bar magnet, torch

21. An electric toaster and a ceiling fan are both running. The toaster mainly uses the _____ effect of current, while the fan mainly uses the _____ effect.

(A) heating; magnetic

(B) magnetic; heating

(C) heating; heating

(D) magnetic; magnetic

22. An electric fuse is best described as a short piece of wire that:

(A) is a very good conductor with a high melting point

(B) increases the current in the circuit (C) stores electric charge for later use

(D) has a low melting point and melts to break the circuit when the current is too high

23. The fuse "protects" the appliances in a circuit. It does this by:

(A) staying solid and carrying any current, however large

(B) glowing to warn the user without breaking the circuit

(C) melting and **breaking** the circuit when the current rises above a safe value

(D) making the current flow faster

24. A house circuit is rated to carry safely up to 5 A. Which fuse should be fitted in it?

(A) a 5 A fuse, so it melts only if the current goes above the safe limit

(B) a 50 A fuse, so it never melts

(C) a 0.5 A fuse, so it melts during normal use

(D) no fuse, to avoid interruptions

25. Too many high-power appliances are switched on at the same socket. The current grows very large and the fuse melts. This situation is an example of:

(A) a short circuit caused by bare wires touching

(B) overloading the circuit

(C) the magnetic effect of current

(D) the cell being connected the wrong way

26. After a fuse has melted ("blown"), the correct action is to:

(A) replace it with a thicker copper wire so it can never blow again

(B) find and remove the fault, then fit a new fuse of the correct rating

(C) join the two melted ends together with tape and reuse it

(D) replace it with a fuse of a much higher rating to avoid trouble

27. A modern device that does the fuse's job but can simply be **switched back on** after it trips, instead of being replaced, is the:

(A) tungsten filament

(B) electromagnet

(C) electric cell

(D) MCB (miniature circuit breaker)

28. Someone replaces a blown 5 A fuse with a thick copper wire that will never melt. The danger of doing this is that:

(A) the bulb will glow more dimly

(B) the circuit can now overheat and catch fire without anything breaking the current

(C) the cells will be charged the wrong way

(D) the magnetic effect will become too strong

- 29.** In 1820, Hans Christian Oersted noticed that a magnetic compass needle placed near a wire **moved** when:
- (A) the wire was simply lying on the table with no current
 - (B) the wire was painted a different colour
 - (C) an electric current was switched on in the wire
 - (D) the wire was made shorter
- 30.** Oersted's experiment showed that an electric current flowing in a wire produces:
- (A) only heat and never any magnetism
 - (B) light only
 - (C) sound but no magnetism
 - (D) a magnetic field around the wire
- 31.** A compass needle lies still below a straight wire. When the current in the wire is switched on, the needle swings aside. If the current is then switched **off**, the needle:
- (A) keeps pointing in the deflected direction for ever
 - (B) melts
 - (C) starts spinning continuously
 - (D) swings back to its original (north–south) direction
- 32.** A current-carrying wire is wound into a coil around a soft-iron nail to make an **electromagnet**. Compared with the straight wire alone, the coil-and-iron arrangement gives:
- (A) a much stronger magnetic effect
 - (B) no magnetic effect at all
 - (C) a weaker magnetic effect
 - (D) only a heating effect, never magnetism
- 33.** Three things are tried on the same electromagnet, one at a time: (i) more turns of wire, (ii) more current, (iii) keeping an iron core inside the coil. Each of these, on its own, makes the electromagnet:
- (A) weaker
 - (B) stronger
 - (C) lose its magnetism
 - (D) reverse into an insulator
- 34.** The biggest reason an electromagnet is more useful than an ordinary permanent bar magnet for lifting and dropping iron scrap is that an electromagnet:
- (A) is always magnetic and can never be switched off
 - (B) works without any electric current
 - (C) is made of plastic
 - (D) can be switched on to lift and switched off to drop the load
- 35.** A crane at a scrap yard uses a large electromagnet. As the operator drives the crane over a mixed heap, the electromagnet will pick up:
- (A) nothing at all, because electromagnets cannot lift things
 - (B) every object in the heap, whatever it is made of
 - (C) only the plastic and rubber objects
 - (D) iron and steel objects, but not the aluminium, copper or plastic
- 36.** In a working electromagnet the soft-iron core is essential because the iron:

- (A) blocks the current completely
- (B) keeps the magnetism for ever, even with no current
- (C) strongly increases the magnetic effect while current flows, then loses it when current stops
- (D) cools the coil down

37. An electromagnet stops working (loses its magnetism) the moment the:

- (A) iron core is painted
- (B) electric current is switched off
- (C) number of turns is increased
- (D) wire is made of copper

38. In an electric bell, the part that becomes a magnet and pulls the iron arm (armature) towards it when current flows is the:

- (A) gong
- (B) electromagnet (coil on the iron core)
- (C) spring
- (D) hammer head

39. In an electric bell, the moment the electromagnet pulls the armature across and the hammer strikes the gong, the contact at the screw is **broken**. The immediate result is that:

- (A) the current stops, the electromagnet loses its magnetism, and the spring pulls the armature back
- (B) the current grows larger and the bell melts
- (C) the electromagnet becomes a permanent magnet
- (D) the bell rings just once and then stops for good

40. An electric bell rings continuously (a buzzing sound) rather than giving a single strike because:

- (A) the contact keeps making and breaking, so the cycle repeats again and again
- (B) the battery voltage keeps rising
- (C) the electromagnet never switches off once started
- (D) the gong is itself a magnet

41. Which single sequence correctly describes one cycle of an electric bell, starting from the switch being pressed?

- (A) hammer hits gong → current stops → electromagnet pulls armature → spring breaks
- (B) current flows → electromagnet pulls armature → hammer hits gong → contact breaks → current stops → spring returns armature → contact remakes
- (C) spring pulls armature → contact breaks → current flows for ever
- (D) gong becomes a magnet → armature melts → bell stops

42. A current-carrying coil with **no** iron core is wound with the same wire and carries the same current as an identical coil **with** a soft-iron core inside it. The coil with the iron core is:

- (A) the stronger magnet
- (B) the weaker magnet
- (C) exactly equal in strength
- (D) not a magnet at all

43. A student wants to make her electromagnet **stronger** without changing the battery. The most sensible single change is to:

- (A) wind many more turns of wire on the same iron core
- (B) remove the iron core
- (C) use fewer turns of wire
- (D) replace the iron core with a wooden stick

44. Which statement about the heating and magnetic effects is correct?

- (A) a current produces only a heating effect, never a magnetic one
- (B) a current produces only a magnetic effect, never a heating one
- (C) a current passing through a wire produces both a heating effect and a magnetic effect at the same time
- (D) heat and magnetism are produced only after the current is switched off

45. Which device makes use of the **magnetic effect** of electric current?

- (A) an electric toaster
- (B) an electric bell
- (C) an immersion water heater
- (D) an incandescent filament bulb

46. A bulb is joined to a single cell and glows. A second identical cell is now added in the **same direction** (+ to -), making a two-cell battery. Compared with one cell, the bulb is now:

- (A) brighter, because the two cells together push a larger current
- (B) dimmer, because two cells share the work
- (C) off, because two cells cancel out
- (D) unchanged, because cells make no difference

47. Two cells are joined positive-to-positive and negative-to-negative (facing against each other) before being put in a circuit with a bulb. The bulb:

- (A) glows extra brightly
- (B) glows with twice the current
- (C) glows normally
- (D) does not glow, because the cells push against each other and cancel out

48. A circuit is drawn as: cell → closed switch → bulb → back to cell. The bulb glows. A student then opens the switch only. The bulb goes out because:

- (A) the cell has been used up
- (B) the switch turned into a fuse
- (C) the bulb's resistance dropped to zero
- (D) the loop is now broken, so no current can flow

49. A nichrome heater coil and a copper connecting wire carry the **same current**. The nichrome coil becomes red-hot while the copper wire stays cool mainly because:

- (A) nichrome has a much higher resistance than copper, so far more heat is produced in it
- (B) copper carries no current
- (C) nichrome is magnetic and copper is not

(D) copper has a lower melting point than nichrome

50. Which of these correctly pairs a circuit component with its job?

(A) switch — to make or break the circuit

(B) bulb — to break the circuit when current is high

(C) fuse — to store charge

(D) cell — to measure the current

51. An immersion rod (used to heat a bucket of water) and an electric bell are compared. They depend, respectively, on the _____ effect and the _____ effect of current.

(A) magnetic; heating

(B) heating; magnetic

(C) heating; heating

(D) magnetic; magnetic

52. A circuit is drawn with the cell symbol's longer line on the left. Current in the external wires is taken to flow out of the:

(A) shorter thick line (negative terminal) round to the longer line

(B) middle of the cell symbol

(C) longer thin line (positive terminal) round to the shorter line

(D) bulb towards the switch only

53. A child claims: "Adding a second cell the right way always makes a bulb dimmer." This claim is:

(A) correct, because more cells slow the current

(B) wrong; correctly adding a cell increases the push and makes the bulb brighter

(C) correct only for tungsten bulbs

(D) wrong; the bulb always goes out

54. The reason a thin fuse wire melts before the thick household wiring overheats is that, for the same large current, the thin fuse wire:

(A) carries no current at all

(B) has a higher resistance and a low melting point, so it heats up fastest and melts first

(C) is a magnet

(D) is a better conductor than the household wire

55. A safety claim says: "A fuse protects a circuit by **conducting** the extra current safely away without ever breaking." This claim is:

(A) correct — that is exactly how a fuse works

(B) wrong — a fuse increases the current instead

(C) correct only for MCBs

(D) wrong — a fuse protects precisely by **melting and breaking** the circuit when the current is too high

56. Which observation would best convince you that a current is flowing in a hidden wire, **without** cutting the wire open?

(A) a nearby compass needle deflects when the supply is switched on

(B) the wire changes colour

(C) the wire becomes heavier

(D) the wire becomes longer

57. A coil electromagnet lifts at most 5 paper clips with one cell. With **two cells** joined correctly (more current) and **more turns** added, the number of paper clips it can lift will most likely:

(A) increase

(B) decrease

(C) stay exactly the same

(D) drop to zero

58. In an electric bell, the role of the **springy metal strip** carrying the armature is to:

(A) keep the electromagnet switched on permanently

(B) pull the armature back so the contact is remade and the cycle repeats

(C) melt when the current is too high (D) measure the voltage of the cell

59. A bulb, a cell and a switch are connected in a single loop, but the bulb does not light even with the switch closed and the cell fresh. A check shows a tiny break in one wire. This break makes the circuit:

(A) a closed circuit that should still light (B) a battery of two cells

(C) a short circuit that lights the bulb brighter

(D) an open circuit, so no current flows and the bulb stays dark

60. Identify the **incorrect** statement.

(A) an electromagnet can be switched on and off, but a permanent bar magnet cannot

(B) an electromagnet will attract iron and steel objects

(C) an electromagnet made stronger by adding turns and current can lift more iron

(D) an electromagnet attracts plastic, aluminium and wood just as strongly as iron

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 14: Electric Current and its Effect

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	A	31	D	41	B	51	B
2	C	12	A	22	D	32	A	42	A	52	C
3	C	13	C	23	C	33	B	43	A	53	B
4	A	14	D	24	A	34	D	44	C	54	B
5	C	15	C	25	B	35	D	45	B	55	D
6	B	16	C	26	B	36	C	46	A	56	A
7	B	17	A	27	D	37	B	47	D	57	A
8	C	18	D	28	B	38	B	48	D	58	B
9	C	19	B	29	C	39	A	49	A	59	D
10	B	20	A	30	D	40	A	50	A	60	D

Answer-key distribution: A = 17, B = 16, C = 14, D = 13.

Worked Solutions

- (A)** In a cell symbol the longer thin line is the positive (+) terminal and the shorter thick line is the negative (–) terminal. So the pair stands for the positive terminal and the negative terminal.
- (C)** Cells form a battery only when joined in a row with the positive terminal of one touching the negative terminal of the next. Joining like-to-like (+ to +) makes them oppose; an air gap breaks the loop.
- (C)** Putting two cells positive-against-positive makes their pushes oppose and cancel. With equal cells the net push is zero, so no current flows and the bulb does not glow.
- (A)** A battery of two cells joined correctly (+ to –) gives a larger total push than one cell, so more current flows and the bulb glows more brightly. (The trap is to think "more cells = dimmer" — that is the misconception in Q53.)
- (C)** A small circle with a cross inside it is the standard symbol for a bulb. (A cell is two parallel lines; a fuse is a line with a small box/loop.)

- 6. (B)** A plain straight line is simply a connecting wire. A cell, a bulb and a switch all have extra marks added to the line.
- 7. (B)** A small gap with a tilted arm lifted away from one contact is the symbol for an **open** switch: the loop is broken there, so current does not flow.
- 8. (C)** The switch in the loop is open, so the circuit is broken and no current can flow. With an open switch the bulb does not glow at all (not "dim", not "fading").
- 9. (C)** A broken wire, a dead cell and a fused filament would all stop the bulb glowing — so each is a possible reason. A **complete closed loop** would let the bulb glow, so it is NOT a reason for failure. The "not a possible reason" answer is the closed-loop option.
- 10. (B)** A bulb glows only when current can flow, and current flows only round a complete closed loop. An open circuit or an open switch stops it; the number of cells is not the essential condition.
- 11. (C)** Four 1.5 V cells joined correctly in a row add up: $4 \times 1.5 = 6$ V. (Dividing instead of adding gives the trap 0.375 V; using just one cell gives 1.5 V.)
- 12. (A)** Same cell and same bulb, but Q gives a dimmer bulb — so less current flows through Q. More opposition to current means higher resistance, so wire Q has the higher resistance. (A perfect insulator or a broken wire would give no glow at all, not a dim one.)
- 13. (C)** Of the choices, only the copper wire is a good conductor, so only it completes the circuit and lets the bulb glow. Wood, rubber and plastic are insulators.
- 14. (D)** Rubber is an insulator: it does not let current pass, so the loop is not completed electrically and the bulb stays dark.
- 15. (C)** Whenever current flows through a wire, the wire warms up — this is the heating effect of current. (Only special high-resistance wires get hot enough to glow; an ordinary wire just warms, but it does heat.)
- 16. (C)** Same current flows in both. The wire with higher resistance turns more electrical energy into heat, so the thin high-resistance wire X gets hotter than the thick low-resistance wire Y.
- 17. (A)** Nichrome is chosen for heater coils because it has a very high resistance (so it produces a lot of heat) and a high melting point (so it can glow red-hot without melting). Copper, being a far better conductor, would barely heat.
- 18. (D)** More current through the same wire produces more heat each second (the heating effect grows with the current). It does not stay the same or fall.

- 19. (B)** Tungsten is used for bulb filaments because of its extremely high melting point: it can be heated white-hot to give light without melting away.
- 20. (A)** Electric iron, toaster and immersion water heater all work by the heating effect (a high-resistance element gets hot). The bell, crane magnet and loudspeaker use the magnetic effect; fan, mixer and pump use motors (magnetic effect too).
- 21. (A)** A toaster heats food using the heating effect of current; a ceiling fan uses a motor, which runs on the magnetic effect of current. In order: heating, then magnetic.
- 22. (D)** A fuse is a short piece of wire with a **low melting point**. When the current is too high it melts and breaks the circuit. It does not store charge or increase current, and it is deliberately NOT a high-melting-point conductor.
- 23. (C)** The fuse protects by **melting and breaking** the circuit when the current rises above a safe value, cutting off the current before wires overheat. It does not safely carry any current, nor merely glow.
- 24. (A)** The fuse rating should match the safe limit, so fit a 5 A fuse: it carries normal current but melts if the current goes above 5 A. A 50 A fuse never protects; a 0.5 A fuse would melt during normal use; no fuse means no protection.
- 25. (B)** Too many high-power appliances on one socket draw a very large total current — this is overloading. (A short circuit is specifically bare live and neutral wires touching, which is a different fault.)
- 26. (B)** The right action is to find and remove the fault, then fit a fresh fuse of the **correct** rating. Using a thicker wire, a much higher rating, or taping the ends all defeat the fuse's protective purpose.
- 27. (D)** A miniature circuit breaker (MCB) does the fuse's job but can simply be switched back on after it trips, instead of being replaced. An electromagnet, a cell and a filament do not protect a circuit this way.
- 28. (B)** A wire that never melts removes the protection: if a fault sends a huge current, nothing breaks the circuit, so the wiring can overheat and start a fire. That is the danger of "fuse-replaced-by-thick-wire".
- 29. (C)** Oersted saw the compass needle move only when an electric **current was switched on** in the wire. A dead wire, a coat of paint, or a shorter length do not deflect the needle.
- 30. (D)** Oersted's experiment showed that an electric current in a wire produces a **magnetic field** around the wire — the magnetic effect of current. It is not "only heat", "sound", or "light only".

- 31. (D)** The deflection lasts only while current flows. Switch the current off and the magnetic effect vanishes, so the needle swings back to its normal north–south direction. It does not stay deflected, spin or melt.
- 32. (A)** Winding the wire into a coil around a soft-iron core concentrates the magnetic effect, giving a much stronger magnet than a single straight wire carrying the same current.
- 33. (B)** Each of more turns, more current, and keeping an iron core inside the coil makes an electromagnet **stronger**. None of them weakens it or destroys its magnetism.
- 34. (D)** The key advantage is control: an electromagnet can be switched on to lift a load and switched off to drop it — a permanent magnet cannot be turned off. (It is not "always magnetic", not plastic, and it does need current.)
- 35. (D)** An electromagnet attracts only magnetic materials — iron and steel — and not aluminium, copper, plastic, etc. So at a scrap yard it picks up the iron and steel but leaves the rest.
- 36. (C)** While current flows, the soft-iron core strongly increases the magnetic effect; when the current stops, soft iron loses its magnetism almost at once. That is exactly why soft iron (not steel) is used and why the magnet can be switched off.
- 37. (B)** An electromagnet's magnetism depends on the current. The moment the current is switched off, it loses its magnetism. Painting the core, adding turns, or using copper wire do not switch it off.
- 38. (B)** In an electric bell the coil wound on the iron core becomes the **electromagnet**; it is the part that becomes a magnet and pulls the iron armature towards it when current flows. The gong, spring and hammer do not become magnets.
- 39. (A)** When the armature is pulled across and the contact at the screw breaks, the circuit opens: the current stops, the electromagnet loses its magnetism, and the spring pulls the armature back. (It does not become permanent, and the current does not grow.)
- 40. (A)** As the armature returns, the contact is remade, current flows again, the electromagnet pulls once more — the contact keeps making and breaking, so the strike repeats again and again, giving a continuous buzzing.
- 41. (B)** One full cycle is: current flows → electromagnet pulls the armature → hammer hits the gong → the contact breaks → current stops → the spring returns the armature → the contact remakes (and it all repeats). The other sequences put the steps out of order.
- 42. (A)** With the same wire and same current, the coil **with** the soft-iron core is the stronger magnet, because the iron core greatly boosts the magnetic effect.

- 43. (A)** Without changing the battery, the most sensible single change to strengthen the electromagnet is to wind many more turns of wire on the same iron core. Removing the core, using fewer turns, or a wooden core would all weaken it.
- 44. (C)** A current flowing through a wire produces **both** a heating effect (the wire warms) and a magnetic effect (a magnetic field forms) at the same time — they are two effects of the one current, not one or the other, and they occur while the current flows.
- 45. (B)** An electric bell works through an electromagnet, so it uses the **magnetic** effect. A toaster, immersion heater and filament bulb all use the heating effect.
- 46. (A)** Adding a second identical cell in the same direction (+ to –) makes a two-cell battery whose pushes add, driving a larger current, so the bulb glows brighter. (Cells "sharing the work" or "cancelling out" are misconceptions.)
- 47. (D)** Joined positive-to-positive and negative-to-negative, the two cells push against each other and cancel out, so effectively no current flows and the bulb does not glow.
- 48. (D)** Opening the switch breaks the loop, so no current can flow and the bulb goes out. (The cell is not used up, and the resistance does not drop to zero; the switch does not become a fuse.)
- 49. (A)** With the same current, the wire of higher resistance produces far more heat. Nichrome has a much higher resistance than copper, so the nichrome coil glows red-hot while the copper wire stays cool. (Copper does carry the current; the difference is resistance, not magnetism or melting point.)
- 50. (A)** The correct pairing is switch — to make or break the circuit. (A bulb gives light, not over-current protection; a fuse breaks the circuit, it does not store charge; a cell supplies energy, it does not measure current.)
- 51. (B)** An immersion rod heats water using the heating effect; an electric bell rings using an electromagnet, which is the magnetic effect. In order: heating, then magnetic.
- 52. (C)** In the external wires, current is taken to flow out of the positive terminal — the longer thin line — round the circuit back to the shorter (negative) line. It does not start from the middle or from the bulb.
- 53. (B)** The claim is wrong. Adding a cell correctly (+ to –) increases the total push, so more current flows and the bulb gets **brighter**, not dimmer. (Connecting it the wrong way is what reduces the push.)
- 54. (B)** For the same large current, the thin fuse wire has a higher resistance and a low melting point, so it heats up the fastest and melts first — before the thicker household wiring can overheat. That is why the fuse is the deliberate weak link.

55. (D) The claim is wrong. A fuse does not conduct the extra current away; it protects precisely by **melting and breaking** the circuit when the current is too high. (It also does not increase the current.)

56. (A) A current produces a magnetic field, which deflects a nearby compass needle. So if a compass needle deflects when the supply is switched on, current is flowing — and you never had to cut the wire. Colour, weight and length do not change with current.

57. (A) More current (from two cells) and more turns both strengthen the electromagnet, so it can lift **more** paper clips than before. The count increases, not decreases.

58. (B) The springy metal strip pulls the armature back once the magnet releases it, so the contact is remade and the make-break cycle repeats. It does not keep the magnet on, melt, or measure voltage.

59. (D) A tiny break in a wire leaves a gap in the loop, making it an **open** circuit: no current flows, so the bulb stays dark even with a closed switch and a fresh cell. (A break is not a closed circuit and not a short circuit.)

60. (D) Statements about switching on/off, attracting iron and steel, and lifting more when made stronger are all true. The **incorrect** statement is that an electromagnet attracts plastic, aluminium and wood as strongly as iron — it attracts only magnetic materials like iron and steel.